Listing and counting systematically

Kevin Hu
November 2026

L1-CO1 Level 1 Foundations · Combinatorics

Count without missing or repeating anything, using organised lists, tree diagrams, the multiplication principle and route-counting on grids.
By the end you can make organised lists that contain every case exactly once; use tree diagrams and the multiplication principle, filling restricted places first; count pairs without double counting and count routes on a grid.
Prerequisites: none. **Contests:** JMC, Junior Kangaroo, JMO, AMC 8. **Time:** about 60 minutes.

1 Lesson

In how many ways can you pay exactly 30p using only 2p, 5p and 10p coins? If you write payments down as they occur to you, you will soon struggle to say whether any are missing, and two lines may be the same payment in a different order. Instead, fix the number of 10p coins first, starting with the largest possible number, then the number of 5p coins. The rest is paid in 2p coins, so the amount left must be even.

- Three 10p coins give 1 way.
- Two 10p coins leave 10p, with zero or two 5p coins. 2 ways.
- One 10p coin leaves 20p, with zero, two or four 5p coins. 3 ways.
- No 10p coins leave 30p, with zero, two, four or six 5p coins. 4 ways.

So there are $1+2+3+4=10$ ways. Nothing is missing, since every payment uses three, two, one or no 10p coins and each line runs through every possible number of 5p coins. Nothing is repeated, since no two payments listed use the same numbers of coins.

1.1 Organised lists

Making an organised list.

1. Decide what one object looks like and how you will record the object (above, how many coins of each kind).
2. Choose one feature and go through its possible values in order, largest to smallest or smallest to largest.
3. For each value, list the objects with that value, again in a fixed order.

Start with the feature that most restricts the rest, as the 10p coins did. If two lines describe the same object, the recording in step 1 was not fixed enough, since $10+5+5+10$ and $10+10+5+5$

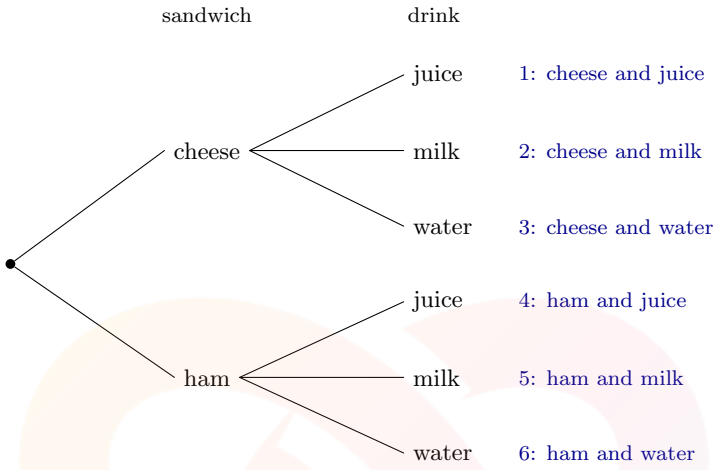
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are the same payment, which is why we recorded how many coins of each kind, not their order.

1.2 Tree diagrams and the multiplication principle

A school canteen offers a cheese or a ham sandwich, and a juice, milk or water. A lunch is one sandwich and one drink. The tree diagram shows every lunch, with two branches for the sandwich and from each of them three branches for the drink.



Each of the 2 sandwich branches splits into 3, so there are $2 \times 3 = 6$ ends, one for each lunch.

Multiplication principle.
If a first choice can be made in a ways, and then, whatever the first choice was, a second choice can be made in b ways, then the two choices together can be made in $a \times b$ ways. The same holds for three or more choices, multiplying the numbers of options.

The *number* of options at a stage must not depend on what was picked earlier, though the options themselves may. Four runners can finish a race in $4 \times 3 \times 2 \times 1 = 24$ orders. Any of the 4 can win, any of the other 3 can be second, either of the remaining 2 third. Which runners can be second depends on the winner, but there are always 3 of them. When one position has a restriction, fill that first (Example 2).

1.3 Counting each object once

Five friends meet and each pair shakes hands once. Each of the 5 shakes 4 hands, giving $5 \times 4 = 20$, but this counts the handshake between Ali and Bea twice, once among Ali's four and once among Bea's, and the same holds for every handshake. So there are $20 \div 2 = 10$. To check, list the pairs with letters in alphabetical order, so each pair is written only one way (AB, AC, AD, AE, BC, BD, BE, CD, CE, DE).

Ask whether swapping two things gives a different object. Swapping the people in a handshake does not. Swapping first and second place in a race does, so there we do not divide.

If a count gives every object exactly twice, divide by 2. More generally, if every object is counted the same number of times k, divide by k. If different objects are counted different numbers of times, dividing does not work, and you must count another way.

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1.4 Paths on a grid

Many problems ask for the number of routes along grid lines from one corner to the opposite corner, moving only right or up. Instead, write at each grid point the number of routes from the start to that point. Every point on the bottom edge or the left edge gets a 1, since the only route there goes straight along the edge. For any other point, the last step of a route arrives either from the point on its left or from the point below, and no route does both. Hence we add.

> routes to a point = routes to the point on its left + routes to the point below.

Fill in the numbers row by row from the bottom and read off the number at the finish.

Example 1.
How many three-digit numbers have digits that add up to 5?

Solution.
Writing down examples (500, 113, 230, ...) gives no way of knowing when we have finished. The hundreds digit is the most restricted, since the digit cannot be 0, so we organise by hundreds, from 5 down to 1, and then take the tens digit from its largest value down to 0. The units digit is then forced.
Hundreds digit 5 (the other two digits add to 0, giving 500). Hundreds digit 4 (the other two add to 1, giving 410 and 401). Hundreds digit 3 (320, 311, 302). Hundreds digit 2 (230, 221, 212, 203). Hundreds digit 1 (140, 131, 122, 113, 104).
The answer is $1+2+3+4+5=15$. Note the pattern. When the last two digits must add up to s (at most 9), the tens digit can be any of $0, 1, \ldots, s$, giving $s+1$ numbers, so the list need not be written in full.

Example 2.
Three-digit numbers are made from the digits 1, 2, 3, 4 and 5, with no digit used more than once. How many of these numbers are odd?

Solution.
Working from left to right, the hundreds digit has 5 options and the tens digit 4, but the number of options for the units digit depends on how many odd digits are already used. Three remain if the first two digits are 2 and 4, but only one if they are 1 and 3. So we cannot multiply in this order.
The restriction is on the units digit, so we choose units first. The units digit must be 1, 3 or 5, giving 3 options. The hundreds digit can then be any of the 4 digits not yet used, and the tens digit any of the remaining 3. The number of options at each stage is now the same whatever was chosen before, so there are $3\times4\times3=36$ odd numbers.
As a check, there are $5\times4\times3=60$ numbers in all and $2\times4\times3=24$ even ones, and $36+24=60$.

Example 3.
At a party of 6 people, each person sends a text message to each of the other five, and each pair of people shakes hands once. How many more messages are sent than handshakes take place?

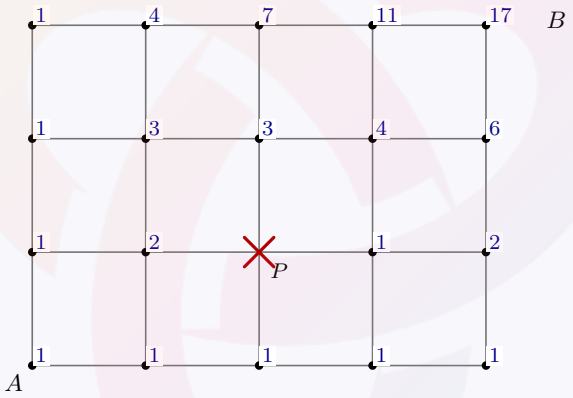
Solution.
Each of the 6 people sends 5 messages. A message from Dan to Eve and a message from Eve

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to Dan are different messages, so nothing has been counted twice, and there are $6 \times 5 = 30$ messages.
For handshakes the same calculation gives 30, but Dan shaking hands with Eve is the same as Eve shaking hands with Dan, so every handshake is counted twice. There are $30 \div 2 = 15$ handshakes, and 15 more messages than handshakes.
In general, n people send $n(n-1)$ messages and make $n(n-1) \div 2$ handshakes. The handshake count also gives the number of games when each of n teams plays each other team once.

Example 4.
An ant walks along the lines of a grid 4 squares wide and 3 squares high, from the bottom left corner A to the top right corner B, always moving right or up. The ant must avoid a drop of glue at P, 2 squares right of A and 1 square up. How many routes are possible?

Solution.
Every route is 4 steps right and 3 up in some order, so listing is slow. We use the labelling method, giving P the number 0, since no allowed route reaches P.



The bottom row and left column are all 1s. The second row is 1, $1+1=2$, then 0 at P, then $0+1=1$ and $1+1=2$. The third row is 1, $1+2=3$, $3+0=3$, $3+1=4$, $4+2=6$, and the top row is 1, $1+3=4$, $4+3=7$, $7+4=11$, $11+6=17$. So there are 17 routes.
To check, count the routes through P and subtract. Without the glue the same method gives 35 routes. There are 3 routes from A to P (the one step up comes first, second or third) and 6 from P to B (RRUU, RURU, RUUR, URRU, URUR, UURR), so by the multiplication principle $3 \times 6 = 18$ routes pass through P, and $35 - 18 = 17$.

Common traps.

- Listing without a rule, then missing a case or writing one twice.
- Counting pairs as if order mattered, or dividing by 2 when order does matter (messages, races, codes).
- Letting a number begin with 0.
- Miscounting a range. There are $40 - 15 + 1 = 26$ whole numbers from 15 to 40 inclusive, not 25.

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- On a grid, multiplying the two neighbouring numbers instead of adding them.

1.5 How to spot this in a contest

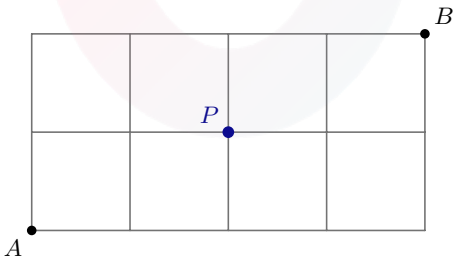
1. "How many" with small answer options (up to about 30). An organised list is usually the fastest safe method.
2. Objects built in stages (digits, sandwich and drink). Multiply the numbers of options, filling restricted positions first.
3. Pairs of people, teams or points (handshakes, games, roads). Count as if order mattered, then divide by 2.
4. Routes that move only in fixed directions. Write the number of routes at every point.
5. Check by a second method where you can, such as counting what you do not want and subtracting.

2 Practice problems

The problems are roughly in order of difficulty, from ⋆ to ⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆ An ice-cream stall serves ice cream in a cone or in a tub, in one of four flavours (vanilla, chocolate, strawberry or mint), with or without a chocolate flake. How many different orders are possible?

Problem 2 ⋆ The grid below is 4 squares wide and 2 squares high. How many routes along the grid lines from A to B, always moving right or up, pass through the point P?



Problem 3 ⋆⋆ Without listing them, explain why there are more four-digit numbers whose digits are all odd than four-digit numbers whose digits are all even.

Problem 4 ⋆⋆ In how many ways can 12 be written as the sum of three positive whole numbers, if the order does not matter? (So $2+4+6$ and $6+2+4$ count as the same way.)

Problem 5 ⋆⋆⋆ Stepping stones in a row are numbered 0 to 8. A frog starts on stone 0 and wants to reach stone 8. Each jump takes the frog forward either 1 stone or 2 stones. Stone 4 is loose, so the frog never lands there. In how many ways can the frog reach stone 8?

Problem 6 ⋆⋆⋆ Using each of the digits 1, 2, 3 and 4 at most once, Nadia writes down every three-digit number that she can make. What is the sum of all the numbers she writes down?

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3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **AMC 8 2004 Problem 2** ⋆
   List the arrangements in order. A number cannot start with 0
2. **UKMT Junior Kangaroo 2022 Question 6** ⋆
   Count all the pairs of cities, then compare with the roads drawn
3. **UKMT Junior Kangaroo 2018 Question 9** ⋆
   Organised list of the two scores, smaller score first. Watch for repeated totals
4. **AMC 8 2022 Problem 3** ⋆
   Organised list by the value of the smallest number
5. **UKMT JMC 2019 Question 13** ⋆⋆
   For each M, count the C squares next to that M
6. **AMC 8 2024 Problem 8** ⋆⋆
   Draw the tree diagram. Different branches can end at the same amount
7. **AMC 8 2020 Problem 7** ⋆⋆
   Decide what the first two digits can be, then list the rest in order
8. **AMC 8 2017 Problem 15** ⋆⋆
   Write the number of ways to reach each letter, working outwards from the centre
9. **UKMT JMC 2026 Question 17** ⋆⋆
   Organised list, branching on each digit in turn
10. **UKMT JMC 2023 Question 19** ⋆⋆⋆
    What does the condition say about any score during the match? Then list in order
11. **AMC 8 2020 Problem 21** ⋆⋆⋆
    Write the number of paths at each white square, row by row
12. **UKMT JMC 2026 Question 25** ⋆⋆⋆
    Split into cases, namely no tie or one pair tied. Multiply inside each case

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4 Hints

1. An order is made of three choices, one after another. Does any choice change the number of options for another?
2. Split a route into the part from A to P and the part from P to B.
3. Count each kind of number with the multiplication principle, one place at a time. Which place behaves differently?
4. Write each sum with its largest number first, and organise your list by the largest number.
5. Work out the number of ways to reach each stone in turn, starting with stone 1. From which stones can the frog arrive at a given stone?
6. How many of Nadia's numbers have 1 as the hundreds digit? Use the same idea for every digit and every place.

5 Solutions

Problem 1

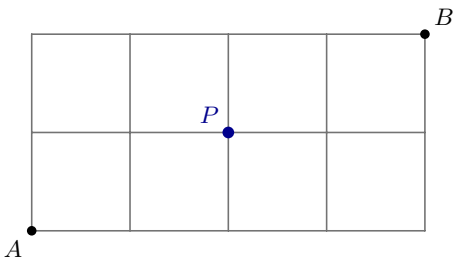
An ice-cream stall serves ice cream in a cone or in a tub, in one of four flavours (vanilla, chocolate, strawberry or mint), with or without a chocolate flake. How many different orders are possible?

An order is a choice of container, then a flavour, then flake or no flake. The container can be chosen in 2 ways, the flavour in 4 ways and the flake in 2 ways, and no choice changes the number of options for the others. By the multiplication principle there are $2 \times 4 \times 2 = 16$ orders.

Answer: 16

Problem 2

The grid below is 4 squares wide and 2 squares high. How many routes along the grid lines from A to B, always moving right or up, pass through the point P?



A route through P is a route from A to P followed by a route from P to B. To get from A to P a route needs 2 steps right and 1 step up, and the step up can come first, second or third (URR, RUR, RRU), so there are 3 such routes. From P to B a route again needs 2 steps right and 1 step up, so there are 3 routes. Any route to P can be followed by any route from P, so by the multiplication principle there are $3 \times 3 = 9$ routes.

Answer: 9

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Problem 3

Without listing them, explain why there are more four-digit numbers whose digits are all odd than four-digit numbers whose digits are all even.

The odd digits are 1, 3, 5, 7 and 9. In a four-digit number with all digits odd, each of the four places can hold any of these five digits, so by the multiplication principle there are $5 \times 5 \times 5 \times 5 = 625$ such numbers.

The even digits are 0, 2, 4, 6 and 8. A four-digit number cannot begin with 0, so the first digit has only 4 options (2, 4, 6 or 8), while each of the other three places has 5. So there are $4 \times 5 \times 5 \times 5 = 500$ four-digit numbers with all digits even.

Since $625 > 500$, there are more numbers of the first kind. The difference comes only from the first digit. Both kinds of number have five digits to choose from, but 0 is even and cannot come first.

Problem 4

In how many ways can 12 be written as the sum of three positive whole numbers, if the order does not matter? (So $2+4+6$ *and* $6+2+4$ *count as the same way.)*

Write each sum as $a+b+c$ with $a \geq b \geq c \geq 1$, so that each way is recorded once. If a were 3 or less, all three numbers would be at most 3 and the total at most 9, so a is at least 4. Since $b+c$ is at least 2, a is at most 10. We list by a, and for each a by b from the largest possible value down.

$a = 10$ gives $10+1+1$. $a = 9$ gives $9+2+1$. $a = 8$ gives $8+3+1$, $8+2+2$. $a = 7$ gives $7+4+1$, $7+3+2$. $a = 6$ gives $6+5+1$, $6+4+2$, $6+3+3$. $a = 5$ gives $5+5+2$, $5+4+3$. $a = 4$ gives $4+4+4$.

This gives $1+1+2+2+3+2+1 = 12$ ways. In each line, stop when b would become smaller than c. For $a = 5$ we have $b+c = 7$, and $b = 3$ would need $c = 4$, which is larger than b (and $5+3+4$ is $5+4+3$, already listed).

Answer: 12

Problem 5

Stepping stones in a row are numbered 0 to 8. A frog starts on stone 0 and wants to reach stone 8. Each jump takes the frog forward either 1 stone or 2 stones. Stone 4 is loose, so the frog never lands there. In how many ways can the frog reach stone 8?

The frog arrives at a stone either by a short jump from the previous stone or by a long jump from the stone two before, and no way of arriving does both. So, as with routes on a grid, the number of ways to reach a stone is the number of ways to reach the previous stone plus the number of ways to reach the stone two before. The frog never lands on stone 4, so stone 4 gets 0.

Stone 0 has 1 way (the frog starts there). Stone 1 has 1. Stone 2 has $1+1 = 2$. Stone 3 has $2+1 = 3$. Stone 4 has 0. Stone 5 has $0+3 = 3$. Stone 6 has $3+0 = 3$. Stone 7 has $3+3 = 6$. Stone 8 has $6+3 = 9$. So there are 9 ways.

As a check, the frog must jump from stone 3 straight to stone 5. There are 3 ways to reach stone 3 (1+1+1, 1+2 and 2+1) and, by the same list, 3 ways to get from stone 5 to stone 8, so there are $3 \times 3 = 9$ ways.

Answer: 9

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Problem 6

Using each of the digits 1, 2, 3 and 4 at most once, Nadia writes down every three-digit number that she can make. What is the sum of all the numbers she writes down?

By the multiplication principle Nadia writes $4 \times 3 \times 2 = 24$ numbers. Those with hundreds digit 1 have their other two digits chosen from 2, 3 and 4, in $3 \times 2 = 6$ ways. In the same way, each of the four digits is the hundreds digit of exactly 6 numbers, and also the tens digit of exactly 6 numbers and the units digit of exactly 6 numbers.

So the hundreds digits of all 24 numbers add up to $6 \times (1 + 2 + 3 + 4) = 60$, and so do the tens digits and the units digits. The sum of the numbers is therefore $60 \times 100 + 60 \times 10 + 60 \times 1 = 6000 + 600 + 60 = 6660$.

Answer: 6660

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